AMIT KUMAR

amitkrbsc26@gmail.com | +91-9953732860 | New Delhi, India | linkedin.com/in/amitkr26

PROFESSIONAL SUMMARY

M.Sc. Electronics graduate from the University of Delhi with hands-on research experience in transition metal thin films and semiconductor devices. Fabricated $Fe_{80}Co_{20}$ thin films via DC sputtering and conducted structural and magnetic characterization using XRD, VSM, and FMR, including extraction of key magnetic parameters. Strong fundamentals in semiconductor physics, digital circuit design, and embedded systems. UGC-NET qualified (Ph.D. eligible). Seeking entry-level opportunities in semiconductor technology, VLSI, hardware design, or R&D

EDUCATION

M.Sc. Electronics, *Department of Electronic Science - University of Delhi South Campus*
2023 – 2025
CGPA: 7.2
Relevant Coursework: Semiconductor Devices, VLSI Technology, Digital Circuit Design, Analog Circuit Design, Embedded Systems, DSP, Fabrication & Characterization Techniques

B.Sc. (Hons.) Electronic Science,
Acharya Narendra Dev College, University of Delhi
2020 – 2023
CGPA: 7.4
Key Subjects: Semiconductor Devices, Signals & Systems, Digital Electronics & VHDL, Microprocessors, Control Systems, Embedded Systems

TECHNICAL SKILLS

Magnetic Characterization
XRD, VSM, FMR

Digital & Embedded
Digital Circuit Design, Verilog (Basic), 8051, Arduino

Programming
C, C++, Python (Basic)

Tools
MATLAB, Proteus, Multisim, PSpice, Keil, Origin

INDUSTRY EXPERIENCE

Technical Support Trainee,
Sony India (via Superwell Services Pvt. Ltd.)
09/2025 – 11/2025
Provided remote troubleshooting for Sony Bravia systems Diagnosed software, display, and connectivity issues Escalated hardware failures to field engineering teams Maintained structured technical documentation

Electronics Intern, *Sagedel Tech LLP*
07/2023 – 12/2023
Designed and tested analog and digital circuits using BJT, 555 Timer IC, and Arduino Built line-following and obstacle-detection robotic system Performed hardware debugging and component validation Assisted in circuit assembly and system verification

RESEARCH EXPERIENCE

M.Sc. Dissertation – Thin Films & Spintronics
01/2025 – 05/2025
Fabricated Fe80Co20 thin films on flexible copper substrate using DC sputtering Performed structural analysis using XRD Conducted magnetic characterization using VSM and FMR Extracted saturation magnetization, coercivity, resonance field, and Gilbert damping constant Studied effect of film thickness on magnetic properties

PROJECTS

ESP32-CAM Based Digital Microscope
Developed live-streaming imaging system using ESP32

Arduino Line Follower Robot
Implemented IR sensor-based path detection and motor control

CERTIFICATIONS

- UGC-NET (Electronic Science) – Qualified for Ph.D. Only
- VLSI for Beginners – NIELIT Calicut